10.24 The Running φ^3 Coupling and the Callan–Symanzik Equation

LecNote13 p. 4 finds the φ^3 vertex "changing with external momentum", LecNote14 p. 8 an amplitude $\propto 1-\frac{11}{12}\alpha\ln(s/m^2)$ (§10.21); LecNote15 runs $\alpha(\mu)$ and stops on p. 16 at the Callan–Symanzik equation. Students ask:

- "Does φ^3 get stronger or weaker at high energy? On LecNote13 p. 4 the vertex log and the propagator log in the margin have opposite signs."
- "The margin says the Z's absorb the $1/\epsilon$ parts and a $\ln k^2$ is left. Why do poles and logarithms come with the same numbers? And nothing fixes κ_C (LecNote13 p. 4): is g arbitrary?"
- "What is γ_φ on LecNote15 p. 16, and what do I do with the equation? Does it cure the $\frac{11}{12}\alpha\ln(s/m^2)$?"

In short: each $1/\epsilon$ comes packaged with $\ln(\mu^2/D)$, so the poles fix the logarithms. V_3 depends on the field's normalization; $V_3[k^2\tilde{\mathbf{\Delta}}]^{3/2}$ does not, and falls like $1-\frac{3}{4}\alpha\ln(k/m)$: $d\alpha/d\ln\mu=-\frac{3}{2}\alpha^2$. Solved at $\mu\sim\sqrt{s}$, the Callan–Symanzik equation splits the amplitude's logarithm into coupling and external legs, $\frac{11}{12}=\frac{3}{4}+\frac{1}{6}$, and sums the leading logs into $[\alpha(\sqrt{s})/\alpha(m)]^{11/9}$. The last subsection does the same for QED, whose charge grows with μ instead.

Conventions are those of §10.14: metric $(-,+,+,+)$, $\tilde{\mathbf{\Delta}}=1/(k^2+m^2-\Pi)$ the exact propagator, $\mathcal{L}_1=\frac{1}{6}Z_g g\varphi^3-\frac{1}{2}A\,\partial\varphi\partial\varphi-\frac{1}{2}Bm^2\varphi^2+Y\varphi$, $A=Z_\varphi-1$, $B=Z_m-1$, $C=Z_g-1$, $\alpha=g^2/(4\pi)^3$; in $d=6-\epsilon$, $g\to g\tilde{\mu}^{\epsilon/2}$ and $\mu^2=4\pi e^{-\gamma}\tilde{\mu}^2$. "On shell" means $\Pi(-m^2)=\Pi'(-m^2)=0$ and $V_3(0,0,0)=g$ (LecNote12–13); $\overline{\text{MS}}$ keeps only the poles, $A=-\alpha/6\epsilon$, $B=-\alpha/\epsilon$ (LecNote15 p. 5), $C=-\alpha/\epsilon$ (p. 9). Page numbers without a LecNote refer to LecNote15; its pages are PDF pages (circled number -8).

Every Pole Carries a Logarithm

With $D=x_1x_3k_1^2+x_2x_3k_2^2+x_1x_2k_3^2+m^2$, Eq. (10.18.1) gives the vertex (LecNote13 p. 3)

$$\frac{V_3}{g}=1+C+\frac{\alpha}{2}\int dF_3\,\Gamma\!\left(\tfrac{\epsilon}{2}\right)\!\left(\frac{4\pi\tilde{\mu}^2}{D}\right)^{\epsilon/2}=1+C+\frac{\alpha}{\epsilon}-\frac{\alpha}{2}\int dF_3\ln\frac{D}{\mu^2}+O(\epsilon).\quad(10.24.1)$$

The scale $\tilde{\mu}$ enters only as $(\tilde{\mu}^2)^{\epsilon/2}$, next to $\Gamma(\epsilon/2)=2/\epsilon-\gamma+\ldots$, and D is the only other scale; so pole and logarithm come as one package, $\frac{2}{\epsilon}+\ln(\mu^2/D)$.

A loop pole of residue r brings $\frac{r}{2}\ln(\mu^2/D)$; the counterterm removes the pole, the logarithm stays, and at large momenta ($D \propto k^2$) the coefficient of $\ln k^2$ is $-\frac{r}{2}$, half the counterterm's residue. For the vertex, $C = -\alpha/\epsilon$ leaves $-\frac{1}{2}\alpha \ln k^2$; for the self-energy, the loop pole $-\frac{\alpha}{6\epsilon}k^2 - \frac{\alpha}{\epsilon}m^2$ (cancelled by $-Ak^2 - Bm^2$) leaves $+\frac{\alpha}{12}k^2 \ln k^2$, Eq. (10.14.6): the "common feature" in the margin of LecNote13 p. 4. The logarithm needs no regulator: in $d = 6$ the loop integral for $V_3(k) - V_3(0)$ converges as it stands (as the one for Π'' does, §10.19) and gives $-\frac{1}{2}\alpha g \int dF_3 \ln(D/m^2)$. The pole is that logarithm read at the ultraviolet end: a cutoff $|\bar{q}| < \Lambda$ gives $\frac{\alpha}{2}\ln(\Lambda^2/D) - \frac{3}{4}\alpha$ under $\int dF_3$, so $\ln \Lambda^2 \leftrightarrow 2/\epsilon + \ln \mu^2$. So the poles alone fix the running (§10.18), and the running is the momentum dependence.

What is "the coupling"?

Finiteness forces only $C \ni -\alpha/\epsilon$; LecNote13 p. 3's $C = -\alpha[1/\epsilon + \ln(\mu/m) + \kappa_C]$ leaves κ_C free, and nothing fixes it. LSZ fixes the mass and the residue, not g (§10.11). Nor is there an on-shell point: three legs with $k_i^2 = -m^2$ and $\sum k_i = 0$ need $k_1 \cdot k_2 = \frac{1}{2}m^2$, but two real on-shell momenta have $|k_1 \cdot k_2| = m^2 \cosh \eta \geq m^2$, η their relative rapidity (a particle cannot decay into two of its own mass). So g is a convention: $\kappa_C = 0$ is $V_3(0,0,0) = g$, and $\kappa_C = -\ln(\mu/m)$ is $\overline{\text{MS}}$, $V_3 = g[1 - \frac{1}{2}\alpha \int dF_3 \ln(D/\mu^2)]$ (LecNote15 p. 9). Shifting κ_C by δ replaces g by $g(1 - \alpha\delta)$, a finite renormalization that drops out once g is fixed by one measurement. Equating the bare couplings of the two schemes gives

$$\alpha_{\text{os}} = \alpha(\mu)\Big[1 + \frac{3}{2}\alpha\Big(\ln\frac{\mu}{m} + K\Big)\Big], \tag{10.24.2}$$

with $K = \int_0^1 dx\, x(1-x)\ln(1 - x + x^2) = \frac{\pi\sqrt{3}}{6} - \frac{17}{18} = -0.0375$ from the on-shell A: the zero-momentum coupling is the $\overline{\text{MS}}$ one at $\mu = m$, up to $-0.056\,\alpha^2$. V_3 is not yet the strength of the interaction: if $\varphi = \zeta^{1/2}\varphi'$, then $\tilde{\mathbf{\Delta}} = \zeta\tilde{\mathbf{\Delta}}'$ and the amputated $V_3 = \zeta^{-3/2}V_3'$. Only $V_3\tilde{\mathbf{\Delta}}^{3/2}$ is the same for both; in a tree of exact pieces (§10.20) each vertex comes with three half-propagators. At the symmetric Euclidean point $k_i^2 = k^2$, the invariant charge is

$$g_{\text{eff}}(k) \equiv V_3\,[k^2\tilde{\mathbf{\Delta}}(k^2)]^{3/2} \simeq g\Big[1 - \Big(\frac{1}{2} - \frac{3}{2}\cdot\frac{1}{12}\Big)\alpha \ln\frac{k^2}{m^2}\Big] = g\Big[1 - \frac{3}{4}\alpha\ln\frac{k}{m}\Big] \tag{10.24.3}$$

up to constants of order α (with masses kept, the three slopes per $\ln k^2$ are $-\frac{1}{2}$, $+\frac{1}{12}$, $-\frac{3}{8}$ to seven digits at $k^2 = 10^8 m^2$). The interaction weakens at high momenta: asymptotic freedom (LecNote15 p. 12). The counterterms in g_{eff} come as $Z_g Z_\varphi^{-3/2}$, whose pole $C - \frac{3}{2}A = -\frac{3}{4}\alpha/\epsilon$ is twice the $-\frac{3}{8}$ per $\ln k^2$; the same factor turns g into the bare coupling, and its pole gives β.

β, γ_m AND γ_φ

Srednicki derives β in (28.13)–(28.22) by the route of LecNote15 pp. 13–15, with the same symbols. The bare $g_0 = Z_\varphi^{-3/2} Z_g g \tilde{\mu}^{\epsilon/2}$ and m_0 do not know μ ((28.3)–(28.5)), while the $\overline{\text{MS}}$ Z's are pure poles whose coefficients depend on α alone (LecNote15 p. 14), so α and m must run. With $\alpha_0 = \tilde{\mu}^\epsilon \alpha\, e^G$, $G \equiv \ln(Z_g^2 Z_\varphi^{-3}) = \sum_n G_n/\epsilon^n$ and $G_1 = 2C_1 - 3A_1 = -\frac{3}{2}\alpha$ (Srednicki's (28.16)), $d\alpha_0/d\ln\mu = 0$ and a $d\alpha/d\ln\mu$ finite at $\epsilon \to 0$ give, power by power in ϵ,

$$\frac{d\alpha}{d\ln\mu} = -\epsilon\alpha + \beta(\alpha), \qquad \beta = \alpha^2 G_1' = -\frac{3}{2}\alpha^2, \tag{10.24.4}$$

or $dg/d\ln\mu = -\frac{3}{4}\alpha g$: §10.18's $(g\,d/dg - 1)a_1$, with $a_1 = (C_1 - \frac{3}{2}A_1)g$ and a factor $\frac{1}{2}$ from $\tilde{\mu}^{\epsilon/2}$. The order-$1/\epsilon$ condition, $G_2' = \alpha G_1'^2$ (LecNote15 p. 15), predicts the two-loop double pole, $G_2 = \frac{9}{8}\alpha^2$: only simple poles carry new information. Integrated, $\alpha(\mu) = \alpha_1/[1 + \frac{3}{2}\alpha_1 \ln(\mu/\mu_1)]$ (p. 11) falls at high μ and blows up at $\mu_1 e^{-2/(3\alpha_1)}$, a scale made from a dimensionless coupling, where perturbation theory fails (p. 12). *Sign convention:* β is LecNote15's $d\alpha/d\ln\mu$; §10.23 writes $d\alpha_s/d\ln\mu^2 = -\beta(\alpha_s)$, in which φ^3 has $\beta = +\frac{3}{4}\alpha^2$, positive like QCD's ($\beta_0 = \frac{3}{4}$ there gives the $\alpha(\mu)$ above). At one loop $d\alpha/d\ln\mu = -\epsilon\alpha$ suffices to differentiate $\ln m = \ln m_0 + \frac{1}{2}\ln(Z_\varphi/Z_m)$ and $\ln Z_\varphi$:

$$\gamma_m \equiv \frac{d\ln m}{d\ln\mu} = -\frac{5}{12}\alpha, \qquad \gamma_\varphi \equiv \frac{1}{2}\frac{d\ln Z_\varphi}{d\ln\mu} = \frac{\alpha}{12}. \tag{10.24.5}$$

LecNote15 gets the same γ_m and β from the μ-independence of $m_{\text{ph}}^2 = m^2[1 + \frac{5}{12}\alpha(\ln(\mu^2/m^2) + c')]$ (pp. 7–8) and of $|T_{\text{obs}}|^2 \propto \alpha^2[1 + 3\alpha\ln\mu + \cdots]$ (p. 11): observables are functions of the bare parameters, which do not know μ.

THE CALLAN–SYMANZIK EQUATION

The bare propagator $\tilde{\mathbf{\Delta}}_0 = Z_\varphi \tilde{\mathbf{\Delta}}$ depends on k, m_0, g_0, ϵ only. With $\tilde{\mathbf{\Delta}}$ a function of k^2, $\alpha(\mu)$, $m(\mu)$ and μ, $d\ln\tilde{\mathbf{\Delta}}_0/d\ln\mu = 0$ is LecNote15 p. 16's

$$\big[\mathcal{D} + 2\gamma_\varphi\big]\tilde{\mathbf{\Delta}} = 0, \qquad \mathcal{D} \equiv \frac{\partial}{\partial\ln\mu} + \beta\frac{\partial}{\partial\alpha} + \gamma_m m\frac{\partial}{\partial m}. \tag{10.24.6}$$

An n-point function of φ gets $+n\gamma_\varphi$; its amputated part Γ_n, divided by n propagators, gets $-n\gamma_\varphi$. *Check:* with LecNote15 p. 5's $\Pi_{\overline{\text{MS}}} = -\frac{\alpha}{12}(k^2 + 6m^2) + \frac{\alpha}{2}\int_0^1 dx\, D\ln(D/\mu^2)$, $D = x(1-x)k^2 + m^2$, one has $\partial\Pi/\partial\ln\mu = -\alpha(\frac{1}{6}k^2 + m^2)$, and at order α the explicit μ, γ_m and γ_φ give $\tilde{\mathbf{\Delta}}^2[-\alpha(\frac{1}{6}k^2 + m^2) + \frac{5}{6}\alpha m^2 + \frac{1}{6}\alpha(k^2 + m^2)] = 0$ ($\beta\,\partial_\alpha$ is order α^2). Without $2\gamma_\varphi$, $-\frac{\alpha}{6}\tilde{\mathbf{\Delta}}$ would be left.

To solve it at $k^2 \gg m^2$, drop m: $\tilde{\mathbf{\Delta}} = k^{-2}F(k^2/\mu^2, \alpha)$, and along the flow $\alpha(\mu)$, F changes only through γ_φ. Run μ to k, where $F(1, \alpha(k)) = 1 + O(\alpha(k))$ has no large logarithm; with $\int \gamma_\varphi \, d\ln\mu = \int \frac{\alpha}{12}\frac{d\alpha}{\beta} = -\frac{1}{18}\ln\alpha$,

$$k^2\tilde{\mathbf{\Delta}} \simeq e^{2\int_\mu^k \gamma_\varphi d\ln\mu'} = \left[\frac{\alpha(\mu)}{\alpha(k)}\right]^{1/9} = 1 + \frac{\alpha}{12}\ln\frac{k^2}{\mu^2} + \cdots , \tag{10.24.7}$$

the logarithm of (10.14.6), now summed. As $\alpha(k) \to 0$ it grows like $(\ln k)^{1/9}$, which (10.15.9) allows.

THE AMPLITUDE: $\frac{11}{12} = \frac{3}{4} + \frac{1}{6}$

In $\overline{\text{MS}}$ the residue R of $\tilde{\mathbf{\Delta}}$ is not 1, and LSZ gives $T = R^2\Gamma_4$ (§10.11). LecNote15 pp. 9–10 find

$$R^{-1} = 1 + \frac{\alpha}{12}\left[\ln\frac{\mu^2}{m^2} + c''\right], \qquad \Gamma_4 = T_0\left[1 - \frac{11}{12}\alpha\ln\frac{s}{\mu^2} + O(\alpha)\right], \tag{10.24.8}$$

with $c'' = (17 - 3\pi\sqrt{3})/3$ and the massless tree $T_0 = -g^2(\frac{1}{s} + \frac{1}{t} + \frac{1}{u})$. The logarithms in Γ_4 are those of §10.21 with μ^2 for m^2: $-\frac{1}{2}\alpha\ln(s/\mu^2)$ per vertex, $+\frac{\alpha}{12}\ln(s/\mu^2)$ for the propagator; at $s \gg m^2$ and fixed angle the remaining $O(\alpha)$ depends on the angle only. Each factor obeys its own equation: $(\mathcal{D} + 2\gamma_\varphi)R = 0$, since $\partial\ln R/\partial\ln\mu = -\frac{\alpha}{6}$; and $(\mathcal{D} - 4\gamma_\varphi)\Gamma_4 = 0$, since $\partial\ln\Gamma_4/\partial\ln\mu = \frac{11}{6}\alpha$ and $\beta\,\partial_\alpha \ln T_0 = -\frac{3}{2}\alpha$ add up to $\frac{1}{3}\alpha = 4\gamma_\varphi$. The product T is μ-independent, as an S-matrix element must be.

So evaluate each factor where it has no large logarithm, Γ_4 at $\mu = \sqrt{s}$ and R at $\mu = m$, and connect them with R's equation, $R(\sqrt{s}) = R(m)\,e^{-2\int_m^{\sqrt{s}}\gamma_\varphi d\ln\mu} = R(m)[\alpha(\sqrt{s})/\alpha(m)]^{1/9}$. With $\alpha \equiv \alpha(m)$, $\alpha(\sqrt{s}) = \alpha/[1 + \frac{3}{4}\alpha\ln(s/m^2)]$ and $T_0 \propto \alpha$,

$$\begin{aligned} T &\simeq T_0\big|_{\alpha(\sqrt{s})}\left[\frac{\alpha(\sqrt{s})}{\alpha}\right]^{2/9} = T_0\big|_\alpha\left[\frac{\alpha(\sqrt{s})}{\alpha}\right]^{11/9} \\ &= T_0\big|_\alpha\left[1 - \frac{11}{12}\alpha\ln\frac{s}{m^2} + \frac{55}{72}\alpha^2\ln^2\frac{s}{m^2} + \cdots\right]. \end{aligned} \tag{10.24.9}$$

- The first term is LecNote14 p. 8's $-\frac{11}{12}$, now made of two numbers from the poles: $-\frac{3}{4}$ from the coupling at $\sqrt{s}$ in T_0 (β), and $-\frac{1}{6} = -\frac{2}{9}\cdot\frac{3}{4}$ from the four external legs (γ_φ). By (10.24.2), $\alpha(m)$ is the on-shell coupling up to $O(\alpha^2)$, which does not touch these logarithms.
- The invariant charge splits it the same way: $V_3^2\tilde{\mathbf{\Delta}} = g_{\text{eff}}^2[k^2\tilde{\mathbf{\Delta}}]^{-2}/k^2$; g_{eff}^2 gives $2(-\frac{3}{8}) = -\frac{3}{4}$ per $\ln s$, and $[k^2\tilde{\mathbf{\Delta}}]^{-2}$ removes the four half-propagators it put on the external legs (exactly 1 on shell): $-\frac{1}{6}$.
- The $\frac{55}{72}$ predicts the two-loop leading log, and (10.24.9) sums all $\alpha^n\ln^n(s/m^2)$. Two-loop β and γ_φ, and the $O(\alpha)$ constants, enter first at $\alpha^n\ln^{n-1}$.

Why $\mu \sim \sqrt{s}$**.** μ is a mass: LecNote15 p. 11's "$\mu = s$" should read $\mu \sim \sqrt{s}$. A one-loop result misses terms $\sim \alpha^2 \ln^2(s/\mu^2)$, small only for $\mu^2 \sim s$; varying μ estimates them (§10.23). In the measured cross section (p. 10), where the legs' $\ln m$ cancels against collinear emission (§10.22), $\mu^2 = s$ leaves $|T_{\rm obs}|^2 = |T_0|^2_{\alpha(\sqrt{s})}[1 - \frac{\alpha}{3}\ln\delta^{-2}]$. So p. 3's $\frac{3}{2}\ln(s/m^2)$, p. 4's "different coefficient", is α^2 running from m to $\sqrt{s}$: $\alpha(\sqrt{s})^2 = \alpha(m)^2[1 - \frac{3}{2}\alpha\ln(s/m^2) + \cdots]$.

QED: the Effective Charge and $\alpha(\mu)$

LecNote23 pp. 4–10 (Srednicki §§62, 66) renormalize QED at one loop and quote $\beta = 2\alpha^2/3\pi$. Students ask: "Where does the Ward identity enter? LecNote23 p. 6 boxes $\Pi'(0) = 0$, but its p. 9 gives a Π with $\Pi'(0) \neq 0$. And $\beta > 0$: what grows, from what, and how far?" The logic is that of φ^3 above, with β of the opposite sign.

The Ward identity, three times. (Srednicki §§67–68, not in the course.) (i) $k_\mu \mathcal{M}^\mu = 0$ (59.15) for an external photon, $\mathcal{T} = \varepsilon_\mu \mathcal{M}^\mu$: the polarization sum becomes $g^{\mu\nu}$, for the full amplitude only (§10.30; LecNote23 p. 3). (ii) $k_\mu \Pi^{\mu\nu} = 0$ ((62.6), (10.20.2)): no photon mass, and the ξ term is never dressed. (iii) $Z_1 = Z_2$ (68.13), so $e_0 = Z_3^{-1/2}\tilde{\mu}^{\epsilon/2}e$: only the photon renormalizes the charge, for every charged field.

One condition. In (62.10), $P_{\mu\nu}/k^2[1 - \Pi(k^2)]$, $k^2\Pi$ plays the role of the Π in $1/(k^2 + m^2 - \Pi)$ above, at $m = 0$. Of the two on-shell conditions, the first (the pole stays at $k^2 = 0$) holds by (ii); the second, $\Pi(0) = 0$ (62.11), fixes Z_3, the photon's only counterterm. So $\Pi'(0)$ is a prediction (LecNote23's note S2); (62.25) gives at $k^2 > 0$

$$\Pi \simeq \frac{\alpha}{15\pi}\frac{k^2}{m^2} \quad (k^2 \ll m^2), \qquad \Pi \simeq \frac{\alpha}{3\pi}\left[\ln\frac{k^2}{m^2} - \frac{5}{3}\right] \quad (k^2 \gg m^2) \tag{10.24.10}$$

($\int_0^1 x^2(1-x)^2 = \frac{1}{30}$, $\int_0^1 x(1-x)\ln[x(1-x)] = -\frac{5}{18}$). Between static charges ($k^2 = \mathbf{k}^2$), $1/[1 - \Pi]$ adds to $q_1q_2/\mathbf{k}^2$ the Uehling term $q_1q_2\Pi/\mathbf{k}^2$. It falls like e^{-2mr}, so on scales $\gg 1/m$ it acts as $q_1q_2\Pi'(0)\delta^3(\mathbf{r})$: in hydrogen it lowers 2S by $\alpha^5 m/30\pi = 1.12 \times 10^{-7}$ eV and hardly moves 2P.

The effective charge. By (iii), $e^2\tilde{\Delta}_{\mu\nu} = \tilde{\mu}^{-\epsilon}e_0^2\tilde{\Delta}_{0\mu\nu}$ depends on neither the scheme nor μ: unlike (10.24.3), QED's invariant charge needs no vertex or legs. Between conserved currents it is $e^2/k^2[1 - \Pi] = 4\pi\alpha_{\rm eff}/k^2$, with

$$\alpha_{\rm eff}(k^2) = \frac{\alpha}{1 - \Pi(k^2)}, \qquad \frac{d\alpha_{\rm eff}}{d\ln k} \xrightarrow{k^2 \gg m^2} \frac{2\alpha_{\rm eff}^2}{3\pi} = \beta(\alpha_{\rm eff}), \tag{10.24.11}$$

(66.10)'s β, as (10.24.1)'s rule predicts from (62.23)'s pole $-\frac{2\alpha}{3\pi\epsilon}$. The charge is screened: from afar it is the on-shell $\alpha = 1/137.036$, and it grows inside $1/m$.

$\alpha(\mu)$ ***and the Landau pole.*** (62.24) and (66.5) agree at $\mu = m$, so equal bare charges give $\alpha(m) = \alpha$ in $\overline{\text{MS}}$, with no K as in (10.24.2). With the electron alone, β integrates to (66.30) for one field (Problem 66.4 adds the others), $1/\alpha(\mu) = 1/\alpha - (2/3\pi)\ln(\mu/m)$, and at $k \gg m$, $\alpha_{\text{eff}}(k^2) = \alpha(e^{-5/6}k)$. The growth is slow, $1/\alpha(\mu) = 134.6$ at $\mu = 10^5 m = 51\,\text{GeV}$, but $\alpha(\mu)$ blows up at the Landau pole $\mu_L = m\, e^{3\pi/2\alpha} \approx 10^{280} m$. It reaches 1 at $\ln(\mu/m) = 641$, where perturbation theory fails.